

FIITJEE
ALL INDIA TEST SERIES
CONCEPT RECAPITULATION TEST – II
JEE (Main)-2022

TEST DATE: 17-02-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

Sol. We must take into account here that the heat transferred per unit time is proportional to the temperature difference. Let us introduce the following notation: T_{out1}, T_{out2} and T_{r1}, T_{r2} are the temperatures outdoors and in the room in the first and second cases respectively. The thermal power dissipated by the radiator in the room is $k_1(T - T_r)$, where k_1 is a certain coefficient. The thermal power dissipated from the room is $k_2(T_r - T_{out})$, where k_2 is another coefficient. In thermal equilibrium, the power dissipated by the radiator is equal to the power dissipated from the room. Therefore, we can write

$$k_1(T - T_{r1}) = k_2(T_{r1} - T_{out1})$$

Similarly, in the second case,

$$k_1(T - T_{r2}) = k_2(T_{r2} - T_{out2})$$

Dividing the first equation by the second, we obtain

$$\frac{T - T_{r1}}{T - T_{r2}} = \frac{T_{r1} - T_{out1}}{T_{r2} - T_{out2}}$$

Hence we can determine T :

$$T = \frac{T_{r2}T_{out1} - T_{r1}T_{out2}}{T_{r2} + T_{out1} - T_{out2} - T_{r1}} = 60^\circ C$$

2. D

 Sol. At time t , the distance travelled by the rod is

$$ED = vt$$

$$\tan \alpha = \frac{AD}{ED} \text{ or } AD = ED \tan \alpha$$

$$AD = vt \tan \alpha$$

$$\tan(90 - \alpha) = \frac{DC}{ED} \text{ or } DC = vt \cot \alpha$$

$$\text{So, } AC = AD + DC = vt(\tan \alpha + \cot \alpha)$$

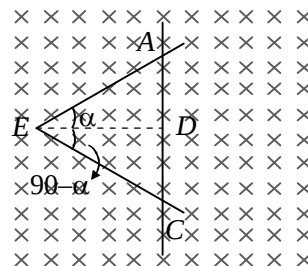
$$\text{Induced emf} = Bvl = Bv(AC)$$

$$= Bv \cdot vt(\tan \alpha + \cot \alpha)$$

$$\text{Induced emf} = Bv^2 t(\tan \alpha + \cot \alpha)$$

 Hence, Induced emf $\propto t$ and v^2

Therefore, choices (B) and (D) are correct and choices (A) and (C) are wrong.



3. A

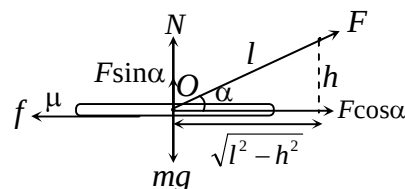
$$\text{Sol. } N = mg - F \sin \alpha$$

$$F \cos \alpha = f = \mu N$$

$$F \cos \alpha = \mu(mg - F \sin \alpha)$$

$$\mu = \frac{F \cos \alpha}{mg - F \sin \alpha} = \frac{F \times \frac{\sqrt{l^2 - h^2}}{l}}{mg - F \times \frac{h}{l}}$$

$$\mu = \frac{F \sqrt{l^2 - h^2}}{mgl - Fh}$$

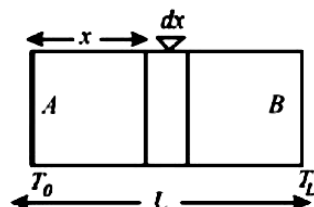


4. A

$$\text{Sol. } C = \sqrt{\frac{\gamma RT}{M_0}} = \sqrt{\frac{5RT}{3M_0}}$$

$$dx = C \cdot dt = \sqrt{\frac{5R}{3M_0}} \left[T_0 + \left(\frac{T_L - T_0}{L} \right) x \right] dt$$

$$t = \frac{2L}{(\sqrt{T_L} + \sqrt{T_0})} \sqrt{\frac{3M}{5R}}$$



5. A

 Sol. Let v_1 and v_2 be the velocity required for the satellite in order to maintain its motion along the circular orbits C_1 and C_2 respectively.

$$\text{Then } v_1 = \sqrt{\frac{GM}{3R}} \text{ and } v_2 = \sqrt{\frac{GM}{6R}}$$

 Now, let v'_1 and v'_2 be the velocities of the satellite at A and B while the transfer is pursued from orbit C_1 to C_2 . Evidently, the points A and B will be the position of perigee and apogee for the satellite along its elliptical path.

Conserving angular momentum

$$v'_1(3R) = v'_2(6R)$$

$$\Rightarrow v'_2 = \frac{v'_1}{2}$$

Conserving energy between A and B, we have

$$\frac{1}{2}m(v'_1)^2 - \frac{GMm}{3R} = \frac{1}{2}m(v'_2)^2 - \frac{GMm}{6R}$$

$$\Rightarrow v'^2_1 - v'^2_2 = 2GM\left(\frac{1}{6R}\right)$$

$$\Rightarrow v'^2_1 - \frac{v'^2_1}{4} = \frac{GM}{3R}$$

$$\Rightarrow \frac{3v'^2_1}{4} = \frac{GM}{3R} \Rightarrow v'_1 = \frac{2}{3}\sqrt{\frac{GM}{R}}$$

$$\text{So, } v'_2 = \frac{1}{3}\sqrt{\frac{GM}{R}}$$

Now, change in energy required at point A

$$\Delta E_1 = \frac{1}{2}mv'^2_1 - \frac{1}{2}mv^2_1$$

$$\Rightarrow \frac{m}{2}\left[\frac{4}{9}(gR) - \frac{gR}{3}\right] \Rightarrow \frac{m}{2}\left(\frac{gR}{2}\right) = \frac{mgR}{18}$$

Similarly change in energy required at point B

$$\Delta E_2 = \frac{1}{2}mv^2_2 - \frac{1}{2}mv'^2_2 = \frac{m}{2}\left[\frac{gR}{6} - \frac{gR}{9}\right]$$

$$= \frac{m}{2}\left(\frac{gR}{18}\right) = \frac{mgR}{36}$$

6. A

Sol. If ring is displaced by an angle θ along x direction then

torque due to magnetic field is $\tau_1 = \mu \times B = IAB\sin\theta = \pi R^2 IB\sin\theta$.

And torque due to spring force is $\tau_2 = K(2R\sin\theta) \times 2R\cos\theta$.

Both torque have the same sense so net torque is $\tau = \tau_1 + \tau_2 = \pi R^2 IB\sin\theta + 4KR^2 \cos\theta \sin\theta$.

For θ to be small $\sin\theta = \theta$

$$\text{So } \tau = -(\pi R^2 IB + 4KR^2)\theta \text{ or } \frac{3MR^2}{2}\alpha = -(\pi R^2 IB + 4KR^2)\theta \text{ or } \alpha = -\frac{2(\pi IB + 4K)}{3M}\theta$$

$$\text{Here } \alpha \propto -\theta \text{ so it is SHM, hence time period is } T = 2\pi\sqrt{\frac{3M}{2(4K + \pi IB)}}$$

7. B

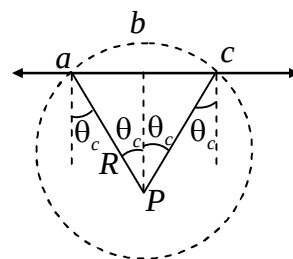
Sol. The light escape is confined within a cone of apex angle ' $2\theta_c$ ' where θ_c is the critical angle. Imagine a sphere with source of light as its centre and the surface area abc is A .

$$\text{here } A = \int_0^{\theta_c} 2\pi R^2 \sin \theta d\theta = 2\pi R^2 (1 - \cos \theta_c)$$

$$= \pi R^2 \left[\because \theta_c = \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) = 60^\circ \right]$$

$$\therefore \text{Power transfer} = P \times \frac{A}{4\pi R^2}$$

$$= 4 \times \frac{1}{4} = 1 \text{ W}$$



8. C

Sol. Thin dark ring will be visible if ray from source get total internal reflection from the spherical shell.

Let the source at any instant be at point P then at point Q ray will be totally reflected if θ is equal to or greater than critical angle. If QP is equal to x , then

$$z = \cos \theta = \frac{R^2 + x^2 - v^2 t^2}{2Rx}$$

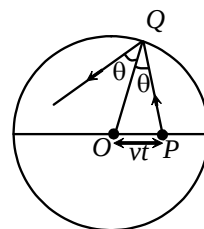
For θ to be maximum

$$\frac{dz}{dx} = \frac{2x(2Rx) - 2R(R^2 + x^2 - v^2 t^2)}{4R^2 x^2} = 0 \Rightarrow x = \sqrt{R^2 - v^2 t^2}$$

$$\text{So, } \cos \theta = \frac{2(R^2 - v^2 t^2)}{2R\sqrt{R^2 - v^2 t^2}} = \frac{\sqrt{R^2 - v^2 t^2}}{R}$$

$$\text{For no light to come out, } \sin \theta \geq \frac{1}{\sqrt{2}} \text{ or } \theta \geq 45^\circ$$

$$\text{For minimum time, } \cos \theta = \frac{1}{\sqrt{2}} \text{ or } \frac{1}{\sqrt{2}} = \frac{\sqrt{R^2 - v^2 t^2}}{R} \text{ or } vt = \frac{R}{\sqrt{2}} \Rightarrow t = \frac{R}{\sqrt{2}v}$$



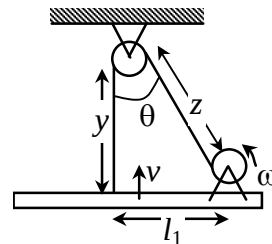
9. A

Sol. $l = y + z$ (total length at any time t)

$$= y + \sqrt{y^2 + l_1^2}$$

$$\frac{dl}{dt} = \frac{dy}{dt} + \frac{y}{\sqrt{y^2 + l_1^2}} \frac{dy}{dt}$$

$$\omega R = v(1 + \cos \theta), \quad v = \frac{\omega R}{1 + \cos \theta}$$



10. A

$$\text{Sol. } \frac{2A}{K^2} \left[Q \log_e \left(\frac{Q - K\sqrt{h_1}}{Q - K\sqrt{h_2}} \right) + K (\sqrt{h_1} - \sqrt{h_2}) \right]$$

11. A

Sol. $F = -\frac{dU}{dr} = -\frac{k}{r}, \frac{k}{r} = \frac{mv^2}{r} \dots (i)$

$$E_n = \frac{1}{2}mv^2 + k \ln r \dots (ii)$$

$$mvr = \frac{nh}{2\pi} \dots (iii)$$

$$\text{Solving these } E_n = \frac{k}{2} \left(1 + \ln \left(\frac{n^2 h^2}{4\pi^2 mk} \right) \right)$$

$$\text{required ratio} = \frac{E_2 - E_1}{E_4 - E_2} = 1.$$

12. B

Sol. $a = g \sin \theta - \mu g \cos \theta$

$$\frac{\ell}{\sin \theta} = \frac{1}{2}at^2 \Rightarrow \left[t = \sqrt{\frac{2\ell}{a \sin \theta}} \right]$$

$$\text{For 't' to be min, } \frac{dt}{d\theta} = 0$$

13. B

Sol. From Newton's third law, force F will act on the block in forward direction

$$\text{Acceleration of block } a_1 = \frac{F}{M}$$

$$\text{retardation of bullet } a_2 = \frac{F}{m}$$

relative retardation of bullet

$$a_r = a_1 + a_2 = \frac{F(M+m)}{Mm}$$

$$\text{Applying } v^2 = u^2 - 2a_r l$$

$$0 = v_0^2 - \frac{2F(M+m)}{Mm} \cdot l \quad \text{or} \quad v_0 = \sqrt{\frac{2Fl(M+m)}{Mm}}$$

Therefore, minimum value of v_0 is

$$\sqrt{\frac{2Fl(M+m)}{Mm}}$$

14. C

Sol. $E_p = \left[\frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} \right] (-\hat{k}) = -\frac{2\sigma}{\epsilon_0} \hat{k}$

15. A

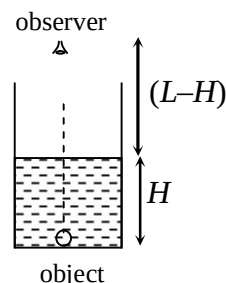
Sol.
$$X_{app} = L - H + \frac{H}{\mu}$$

$$\frac{dX_{app}}{dt} = \frac{dH}{dt} \left(\frac{1-\mu}{\mu} \right)$$

$$\pi r^2 H = V$$

$$\therefore \frac{dH}{dt} = -\frac{2H}{r} \frac{dr}{dt} = \frac{2KH}{r}$$

$$\therefore \frac{dX_{app}}{dt} = \frac{2KV}{\pi r^3} \left(\frac{(1-\mu)}{\mu} \right)$$



16. C

 Sol. Let speed of the bullet = v

 Speed of the system after the collision = V

 By conservation of momentum $mv = (m + M)V$

$$\Rightarrow V = \frac{mv}{M + m}$$

So the initial K.E. acquired by the system

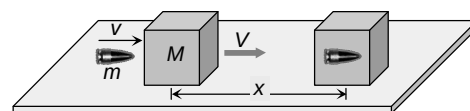
$$= \frac{1}{2}(M + m)V^2 = \frac{1}{2}(m + M) \left(\frac{mv}{M + m} \right)^2 = \frac{1}{2} \frac{m^2 v^2}{(m + M)}$$

 This kinetic energy goes against friction work done by friction = $\mu R \times x = \mu(m + M)g \times x$

By the law of conservation of energy

$$\frac{1}{2} \frac{m^2 v^2}{(m + M)} = \mu(m + M)g \times x \Rightarrow v^2 = 2\mu g x \left(\frac{m + M}{m} \right)^2$$

$$\therefore v = \sqrt{2\mu g x} \left(\frac{M + m}{m} \right)$$



17. C

 Sol. Let at any instant separation between plates of capacitor B is x , then

$$C_{eq} = \frac{\frac{a\epsilon_0}{d} \cdot \frac{a\epsilon_0}{x}}{\frac{a\epsilon_0}{d} + \frac{a\epsilon_0}{x}} = \frac{a\epsilon_0 (dx)}{dx (d + x)} = \frac{a\epsilon_0}{d + x}$$

$$Q = C_{eq} V = \frac{a\epsilon_0}{d + x} V$$

$$\frac{dQ}{dt} = -\frac{a\epsilon_0}{(d + x)^2} V \frac{dx}{dt} = -\frac{a\epsilon_0}{(d + x)^2} V v$$

$$\text{Rate of work done on the battery} = -\left(\frac{dQ}{dt} \right) V = \frac{a\epsilon_0 v V^2}{9d^2} \quad (\text{when } x = 2d)$$

18. A

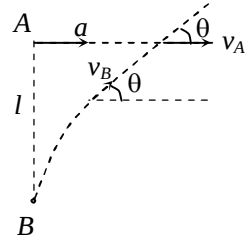
Sol. Let after time t , the velocity of particle B is directed at an angle θ with the horizontal, then

$$-\frac{ds}{dt} = bt - at \cos \theta$$

$$\Rightarrow -\int_l^0 ds = b \int_0^t t dt - a \int_0^t t \cos \theta dt$$

$$\text{and } \frac{1}{2} at^2 = b \int_0^t t \cos \theta dt \quad \therefore l = \frac{bt^2}{2} - \frac{a^2 t^2}{2b}$$

$$t = \sqrt{\frac{2bl}{b^2 - a^2}}, \quad S = \frac{1}{2} bt^2 = \frac{1}{2} b \frac{2bl}{b^2 - a^2} = \frac{b^2 l}{b^2 - a^2}$$



19. B

Sol. Let r is the internal resistance of cell

$$\text{Case I: current in loop } I_1 = \frac{\varepsilon}{R_1 + r}$$

$$\therefore H_1 = I_1^2 R_1 = \frac{\varepsilon^2}{(R_1 + r)^2} \times R_1$$

$$\text{Case II: Current in loop } I_2 = \frac{\varepsilon}{R_2 + r}$$

$$\therefore H_2 = I_2^2 R_2$$

$$\therefore \frac{\varepsilon^2}{(R_2 + r)^2} \times R_2$$

$$\text{i.e., } \frac{\varepsilon^2 R_1}{(R_1 + r)^2} = \frac{\varepsilon^2 R_2}{(R_2 + r)^2}$$

$$\therefore \sqrt{\frac{R_1}{R_2}} = \frac{R_1 + r}{R_2 + r}$$

$$\therefore r = \sqrt{R_1 R_2}$$

20. A

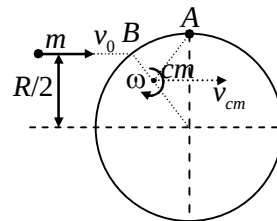
Sol. COM, $mv_0 \hat{i} = 2m\vec{v}_{cm} \Rightarrow \vec{v}_{cm} = \frac{v_0}{2} \hat{i}$

$$\text{COAM about CM } \left(mv_0 \frac{R}{2} \sin 30^\circ \right) (-\hat{k}) = \left(m \frac{R^2}{4} + \frac{1}{2} m R^2 + \frac{m R^2}{4} \right) \vec{\omega}$$

$$\vec{\omega} = \frac{v_0}{4R} (-\hat{k})$$

$$\vec{v}_A = \vec{v}_{cm} + \vec{\omega} \times \vec{r}$$

$$= \frac{v_0}{2} \hat{i} + \frac{v_0}{4R} (-\hat{k}) \times \left(\frac{R}{2} \cdot \frac{\sqrt{3}}{2} \hat{i} + \frac{3R}{4} \hat{j} \right)$$



$$= \frac{11v_0}{16} \hat{i} - \frac{\sqrt{3}v_0}{16} \hat{j}$$

$$|\vec{v}_A| = \frac{\sqrt{31}}{8} v_0$$

SECTION – B

21. 00004.20

Sol. When an excited atom emits a photon of momentum p , conservation of momentum requires that atom recoil with same momentum p .
For emitted photon,

$$\Delta E = \frac{hc}{\lambda} = pc$$

$$\therefore \frac{hc}{\lambda} = p$$

Here, $\Delta E = E_5 - E_1$

$$= \left(-\frac{13.6}{25} \right) - \left(-\frac{13.6}{1} \right)$$

$$= 13.6 \times \frac{24}{25} \times 1.6 \times 10^{-19} \text{ J}$$

Momentum of photon,

$$p = \frac{13.6 \times 24 \times 1.6 \times 10^{-19}}{25 \times 3 \times 10^8}$$

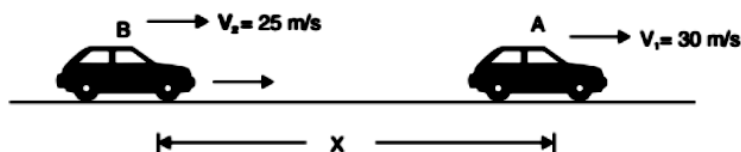
$$= 6.96 \times 10^{-27} \text{ kg ms}^{-1}$$

So, recoil speed of H-atom is

$$\frac{p}{m} = \frac{7 \times 10^{-27}}{1.67 \times 10^{-27}} = 4.20 \text{ ms}^{-1}$$

22. 00023.33

Sol.



Let x = distance between the car at the instant first bullet is fired (sat at time $t = 0$)

Speed of bullet, relative to ground is

$$V_b = u + V_2 = 305 + 25 = 330 \text{ m/s}$$

Velocity of bullet, relative to car A is

$$V_{bA} = 330 - 30 = 300 \text{ m/s}$$

$\therefore$ Time when the first bullet hits the car A is

$$t_1 = \frac{x}{V_{bA}} = \frac{x}{300} \quad \dots (i)$$

Distance between the car when the last bullet is fired (at time $t = T_0 = 20 \text{ s}$) is

$$x^1 = x + V_{AB} T_0 = x + 5 \times 20 = x + 100$$

Time when the last bullet hits the car A is

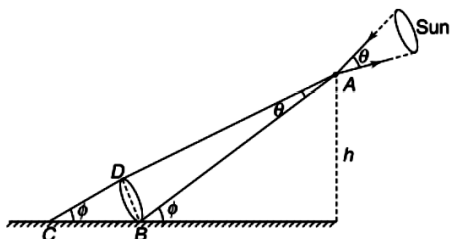
$$t_2 = T_0 + \frac{x}{V_{bA}}$$

$$= 20 + \frac{x + 100}{300} \quad \dots (ii)$$

The interval $\Delta t = t_2 - t_1$
 $= 20 + \frac{100}{30} = 23.33 \text{ sec.}$

23. 00003.44

Sol. The cone of rays passing through the hole at A produce an elliptical spot on the floor. The circular base having diameter BD will get projected on the floor as an ellipse.



CB = major axis = 12 cm

DB = Minor axis = 6 cm

$$DB \approx AB(\theta) = \frac{h\theta}{\sin \phi} \Rightarrow h = (6 \text{ cm}) \frac{\sin \phi}{\theta}$$

But $\sin \phi = \frac{DB}{CB} = \frac{6}{12} = \frac{1}{2}$

$$\therefore h = \frac{(6 \text{ cm})}{2 \times \theta} = \frac{3 \text{ cm} \times 180}{0.5 \times \pi} \left[0.5^\circ = \frac{0.5 \times \pi}{180} \text{ radian} \right]$$

24. 00005.00

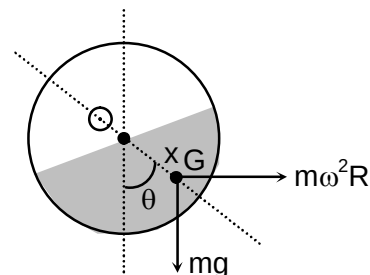
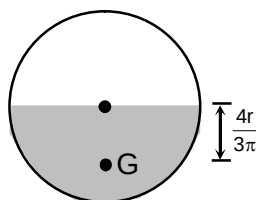
Sol. $I\alpha = -mgx \sin \theta$

$$\Rightarrow \frac{1}{2}mr^2 \frac{d^2\theta}{dt^2} = -mgx \sin \theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{2g}{r^2} \left(\frac{4r}{3\pi} \right) \theta$$

$$\therefore \omega = \sqrt{\frac{8g}{3\pi r}} \Rightarrow f$$

$$= 0.46 \text{ Hz}$$



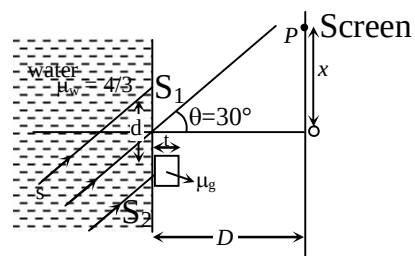
25. 00001.66

Sol. If central maxima is formed at point P on the screen, then

$$\text{Path difference} = S_2P + (\mu_g - 1)t - S_1P - \mu_w d \sin 30^\circ = 0$$

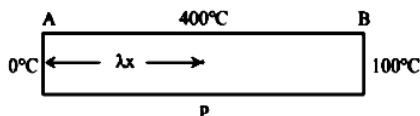
$$\Rightarrow S_2P - S_1P = \frac{xd}{D} = \mu_w d \sin 30^\circ - (\mu_g - 1) \times t$$

$$\Rightarrow x = -1.66 \text{ cm}$$



26. 00009.00

Sol.



For heat flow from P to 0

$$L_f \frac{dm_1}{dt} = \frac{KA400}{\lambda x} \quad \dots(i)$$

For heat flow from P to B

$$L_{\text{vap}} \frac{dm_2}{dt} = \frac{KA300}{10x - \lambda x} \quad \dots(ii) \left[\text{Given } \frac{dm_1}{dt} = \frac{dm_2}{dt} \right]$$

On solving (i) and (ii), we get $\lambda = 9$

27. 00012.50

Sol. Circuit can be redrawn as figure (A)

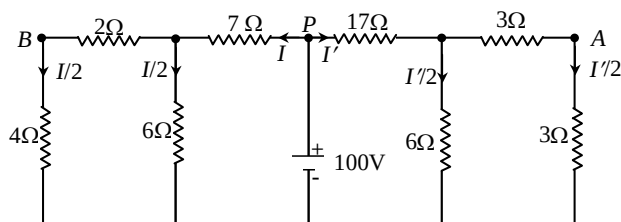


Figure (A)

It is further reduced using series and parallel combination of resistances and is shown in figure (B)

And consequently in figure (C)

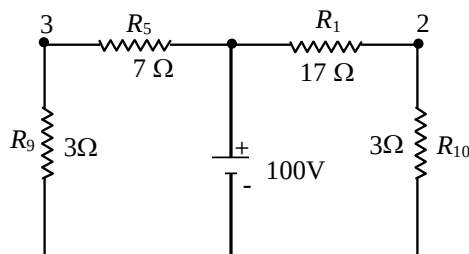


Figure (B)

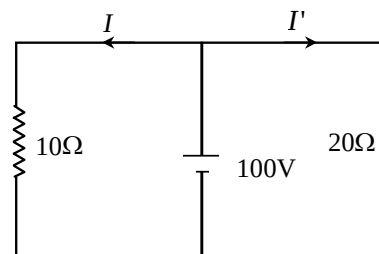


Figure (C)

Current in the $10\ \Omega$ resistor is $I = \frac{100}{10} = 10\ \text{A}$ and in the $20\ \Omega$ resistor is $I' = \frac{100}{20} = 5\ \text{A}$

As shown in figure (A) potential difference between points P and B is

$$V_P - V_B = 10 \times 7 + 5 \times 2 = 80\text{V} \quad \dots (i)$$

Also potential difference between points P and A is

$$V_P - V_A = 5 \times 17 + 2.5 \times 3 = 92.5\ \text{V} \quad \dots (ii)$$

Subtracting (i) and (ii) we get, $V_B - V_A = 12.5\ \text{V}$

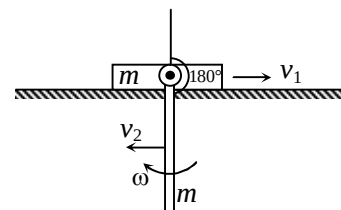
28. 00010.00

Sol. There is no horizontal force, momentum is conserved

$$mv_1 - mv_2 = 0$$

For hinged point $v_1 = \frac{L}{2}\omega - v_2$

$$\text{Energy conservation, } mgL = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}\frac{mL^2\omega^2}{12}$$



Solving, $v_1 = \sqrt{\frac{3}{5}gL}$, $v_1 = 10 \text{ m/s}$.

29. 00020.00

Sol. $\vec{V}_{\text{obj, mirror}} = 4\hat{i} + 9\hat{j}$

$$\frac{dx}{dt} = 4, \frac{dy}{dt} = 9$$

$$u = -x$$

$$-\frac{1}{10} = \frac{1}{V} + \frac{1}{-x}$$

$$V = \frac{-10x}{x-10}$$

$$V_{1x} = \frac{dV}{dt} = \left(\frac{10}{x-10}\right)^2 \left(\frac{dx}{dt}\right)$$

$$V_{1x} = -16$$

$$m = -\frac{V}{-x} = \frac{-10}{x-10} = \frac{y_1^0}{y}$$

$$V_{1y} = -\left(\frac{10}{x-10}\right) \frac{dy}{dt}$$

$$V_{1y} = -18$$

$$\vec{V}_{\text{image, mirror}} = -16\hat{i} - 18\hat{j}$$

$$\vec{V}_{\text{image}} = -12\hat{i} - 16\hat{j}$$

$$|V| = 20 \text{ cm/s}$$

30. 00006.40

Sol. Let R be the radius of the circular layer of water.

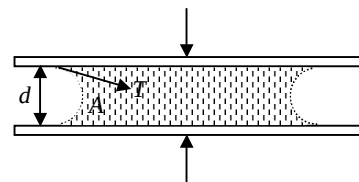
Then, $\pi R^2 d \times \rho = m$... (1)

Pressure at $A = p_0 - \frac{2T}{d}$... (2)

Thus pressure between the plates is less than the atmospheric pressure and so the plates are pressed together as though attracted towards each other.

$$F, \text{ force of attraction} = \Delta p \times \text{area} \Rightarrow F = \frac{2T}{d} \times \pi R^2$$

$$\Rightarrow F = \frac{2T}{d} \times \frac{m}{d \cdot \rho} = \frac{2Tm}{d^2 \cdot \rho} = \frac{2 \times 0.4 \times 10^{-3} \times 0.08}{0.01^2 \times 10^{-4} \times 1000} = 6.4 \text{ N}$$

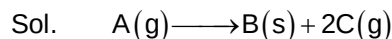


Chemistry

PART – B

SECTION – A

31. C



$$t_{1/2} = 20 \text{ min}$$

$$\text{No. of half lives} = \frac{60}{20} = 3$$

$$\text{Pressure of A after 60 min} = \frac{200}{2^n}$$

$$= \frac{200}{2^3} = 25 \text{ mm}$$

$$\text{Pressure of C after 60 min} = 175 \times 2 = 350 \text{ mm}$$

$$\text{Total pressure} = P_A + P_C = 25 + 350 = 375$$

32. B

Sol. $K = Ae^{-E_a/RT}$

$$E_a = \text{Threshold energy} - \text{energy of reactant} = 90.5$$

$$\log K = \log A - \frac{E_a}{2.303RT}$$

$$\log \frac{0.693}{38.5 \times 60} = 12.3 - \frac{90500}{2.303 \times 8.314 \times T}$$

$$\log 3 \times 10^{-4} 12.3 - \frac{4728.55}{T}$$

$$T = \frac{4276.55}{15.8} = 270$$

33. A

Sol. Application of Le Chatelier's principle.

34. B

Sol. At greatest buffer capacity $[\text{salt}] = [\text{undissociated acid}]$

35. B

Sol. The wavelengths are in the following order

$$4 \rightarrow 1 > 3 \rightarrow 1 > 2 \rightarrow 1 > 3 \rightarrow 2$$

Violet end red end

So the second line from the red end will be corresponding to transition $2 \rightarrow 1$.

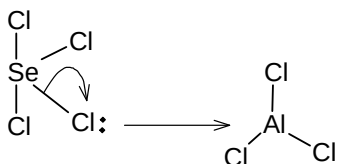
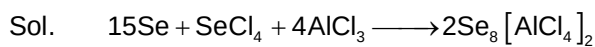
36. D

Sol. No species contains exclusively 90° bond angles. In SF_6 diagonally opposite $F - S - F$ bond angles are at 180° .

37. D

Sol. All the factors given options (A), (B) and (C) are responsible for the hardness of transition metals.

38. C



39. A

Sol. Since an oxygen is added, carboxylic acid must form, so X must be primary alcohol

40. B

Sol. Bulky strong base will cause elimination in other cases

41. D

Sol. Haloform reaction.

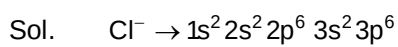
42. C

Sol. One salt will contribute phenyl and other will contribute methyl with C coming from either

43. D

Sol. Factual

44. D



Total electron = 12

45. C

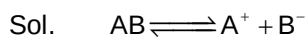
Sol. $U_t = \sqrt{\frac{3RT}{M}} = \frac{5 \times 10^4}{10 \times 10^4}$

$$\frac{\sqrt{\frac{3RT_1}{M}}}{\sqrt{\frac{3RT_2}{M}}} = \frac{1}{4} = \frac{T_1}{T_2}$$

$$T_2 = 4T_1$$

$$T = 4 \text{ times } T_1$$

46. D



$$s = \sqrt{K_{sp}} = 2 \times 10^{-5} \text{ moles/lit}$$

$\therefore$ in 5 litres of water $5 \times (2 \times 10^{-5})$ moles of AB will dissolve.

$$= 10^{-4} \text{ moles}$$

$$= 10^{-4} \times 100 \text{ g}$$

$$= 0.01 \text{ g}$$

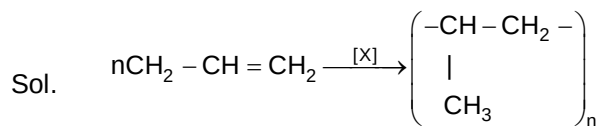
47. D

Sol. In AB process, volume does not change hence it is isochoric process.

48. C

 Sol. Cl_2O_6 when dissolves in water produces HClO_3 and HClO_4 , therefore, it is mixed anhydride.

49. D

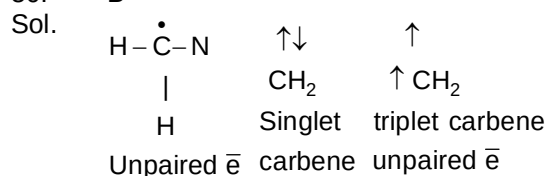


↓

Reagent: 1. Trimethyl aluminum & titanium tetrachloride

2. Zeigler natta catalyst

50. B



SECTION – B

51. 00002.00

Sol. At time = 0

The radio, activity of both is same.

$$\lambda_1 N_1 = \lambda_2 N_2$$

$$\lambda_A N_A = \lambda_B N_B$$

$$\frac{\lambda_A}{\lambda_B} = \frac{N_B}{N_A}$$

$$\therefore N_A : N_B = 2 : 1$$

52. 00000.66

$$\text{Sol. } \frac{4x^2}{(1-x)(0.5-x)} = K_c = 4$$

53. 00033.00

$$\text{Sol. } \frac{x^2}{(a-x)^2} = 4$$

54. 00099.90

$$\text{Sol. } [\text{Ag}^+] \text{ required for ppt. of AgCl} = \frac{1 \times 10^{-10}}{0.1} = 1 \times 10^{-9} \text{ M}$$

$$[\text{Br}^-] \text{ remaining when AgCl starts precipitating} = \frac{1 \times 10^{-13}}{1 \times 10^{-9}} = 10^{-4}$$

$$\% \text{ Br}^- \text{ remaining unprecipitated} = \frac{1 \times 10^{-4}}{0.1} \times 100 = 0.1$$

$$\% \text{ Br}^- \text{ precipitated} = 100 - 0.1 = 99.9$$

55. 00011.30

$$\text{Sol. } [\text{OH}^-] = \frac{3 \times 10^{-2} - 2 \times 10^{-2}}{5} = 2 \times 10^{-3} \text{ M}$$

$$\text{pOH} = -\log(2 \times 10^{-3}) = 2.7$$

$$\text{pH} = 14 - 2.7 = 11.3$$

56. 00000.45

$$\text{Sol. } \frac{\frac{2.45}{98} + \frac{280}{100} \times 7 \times \frac{1}{98}}{500} \times 1000 = \frac{0.025 + 0.20}{500} \times 1000$$

57. 00150.00

Sol. The mass of O atoms = 1.16 – 1.00 = 0.160 g

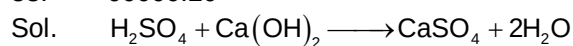
$$\text{Number of moles of O atom} = \frac{0.160}{16} = 0.01$$

$$\text{In } \text{X}_2\text{O}_3 \frac{\text{Number of moles X}}{\text{Number of moles of O}} = \frac{2}{3}$$

$$\text{Number of moles of X} = \frac{2}{3} \times 0.01 = \frac{0.02}{3} = 0.00666$$

$$\text{Atomic weight of X} = \frac{\text{mass in g}}{\text{No. of moles}} = \frac{1.0}{0.00666} = 150$$

58. 00000.20



Initially 0.5mole 0.2mole – –

Finally 0.3mole 0 0.2mole 0.4mole

Limiting reagent is $\text{Ca}(\text{OH})_2$

59. 00002.85

$$\text{Sol. } \frac{r_c}{r_a} = \frac{1.22}{1.25} = 0.976$$

Hence Ga has in body centred or cubic void where $\sqrt{3}a = 2(r_c + r_a)$

$$\sqrt{3}a = 2 \times [1.25 + 1.22] = 4.94$$

60. 00335.00

Sol. For a bcc lattice, $2(r^+ + r^-) = \sqrt{3}a$

Where r = radius of cation

r = radius of anion

a = edge length

$$\therefore r^+ + r^- = \frac{\sqrt{3} \times 387}{2} = 335.142 \text{ pm} \approx 335 \text{ pm}$$

Mathematics

PART – C

SECTION – A

61. B

Sol. Given $\frac{2a+3c}{b+c} + 3 = 0$

$$\Rightarrow 2a + 3b + 6c = 0.$$

$$\text{Now } \int_0^1 (ax^2 + bx + c) dx = \left[\frac{ax^3}{3} + \frac{bx^2}{2} + cx \right]_0^1 = \frac{1}{6} [2a + 3b + 6c] = 0.$$

$\Rightarrow$ atleast one root of $ax^2 + bx + c = 0$ lie in $(0, 1)$.

62. C

Sol. $\cos^{-1}(\log_3(x^2 + 17x + 75))$ is defined if $-1 \leq \log_3(x^2 + 17x + 75) \leq 1$.

$$\Rightarrow \frac{1}{3} \leq x^2 + 17x + 75 \leq 3.$$

$$\text{Now, } x^2 + 17x + 75 \leq 3$$

$$\Rightarrow x^2 + 17x + 72 \leq 0$$

$$\Rightarrow (x + 8)(x + 9) \leq 0$$

$$\Rightarrow x \in [-9, -8]$$

$$\text{and } x^2 + 17x + 75 \geq \frac{1}{3}.$$

$$\Rightarrow 3x^2 + 51x + 225 - 1 \geq 0$$

$$\Rightarrow 3x^2 + 51x + 224 \geq 0$$

Here $D < 0$.

$$\Rightarrow 3x^2 + 15x + 224 > 0 \quad \forall x$$

Hence $x \in [-9, -8]$.

63. C

$$\begin{aligned} \text{Sol. } \lim_{t \rightarrow 0} \frac{\int_0^t \ln(1+x) dx}{t^2} \left(\frac{0}{0} \right) &= \lim_{t \rightarrow 0} \left(\frac{\ln(1+t) + \ln(1-t)}{2t} \right) \\ &= \frac{1}{2} \lim_{t \rightarrow 0} \left(\frac{\ln(1+t)}{t} + \frac{\ln(1-t)}{t} \right) = \frac{1}{2} (1-1) = 0. \end{aligned}$$

64. D

$$\begin{aligned} \text{Sol. } \int \frac{\operatorname{cosec}^2 x - 2005}{\cos^{2005} x} dx &= \int (\cos x)^{-2005} \cdot \operatorname{cosec}^2 x dx - 2005 \int \frac{dx}{\cos^{2005} x} \\ &= (\cos x)^{-2005} \cdot (-\cot x) - \int (-2005)(\cos x)^{-2006} (-\sin x)(-\cot x) dx - 2005 \int \frac{dx}{\cos^{2005} x} \\ &= -\frac{\cot x}{(\cos x)^{2005}} + C. \end{aligned}$$

65. C

Sol. $\alpha + i\beta = \left(\frac{-1+i\sqrt{3}}{2} \right)^{3n_1/4} (1-i)^{-2n_2}$.

Here $\frac{-1+i\sqrt{3}}{2} = e^{\frac{i2\pi}{3}}$ and $1-i = \sqrt{2} e^{\frac{-i\pi}{4}}$

Now, $\alpha + i\beta = e^{\frac{i\pi n_1}{2}} \cdot (\sqrt{2})^{-2n_2} e^{\frac{i\pi n_2}{2}}$

$\Rightarrow \alpha + i\beta = 2^{-n_2} e^{i(n_1+n_2)\pi/2}$

$\Rightarrow \alpha = 2^{-n_2} \cos(n_1+n_2) \frac{\pi}{2} = 0$, when $n_1 + n_2$ odd.

$\beta = 2^{-n_2} \sin(n_1+n_2) \frac{\pi}{2} = 0$, when $n_1 + n_2$ even.

Thus (C) is false.

66. A

Sol. Assuming $\arg z_1 = \theta_1$ and $\arg z_2 = \theta_1 + \theta_2$

$$\frac{\alpha\beta z_1}{\gamma\delta z_2} + \frac{\gamma\delta z_2}{\alpha\beta z_1} = \frac{\alpha\beta |z_1| e^{i\theta_1}}{\gamma\delta |z_2| e^{i(\theta_1+\theta_2)}} + \frac{\gamma\delta |z_2| e^{i(\theta_1+\theta_2)}}{\alpha\beta |z_1| e^{i\theta_1}}$$

$$= e^{-i\theta_2} + e^{i\theta_2} = 2 \cos \theta_2$$

which lies in $[-2, 2]$.

67. D

Sol. Re-writing the equation

$(y-2)^2 = 8\left(x - \frac{1}{3}\right)$, which is parabola with vertex $\left(\frac{1}{3}, 2\right)$ and axis along the line $y = 2$ and

focus is $\left(\frac{7}{3}, 2\right)$.

68. A

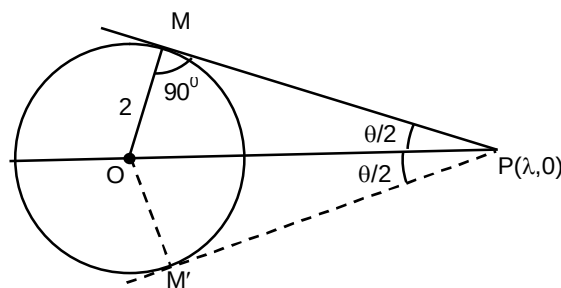
Sol. We have $\frac{\pi}{2} < \theta < \frac{2\pi}{3}$

i.e. $\frac{\pi}{4} < \frac{\theta}{2} < \frac{\pi}{3}$.

$$\Rightarrow \frac{1}{\sqrt{2}} < \sin \frac{\theta}{2} < \frac{\sqrt{3}}{2}$$

But $\sin \frac{\theta}{2} = \frac{2}{\lambda} \Rightarrow \frac{1}{\sqrt{2}} < \frac{2}{\lambda} < \frac{\sqrt{3}}{2}$

$$\Rightarrow \frac{4}{\sqrt{3}} < \lambda < 2\sqrt{2}.$$



69. A

Sol. $x^2 - 2(4a-1)x + 15a^2 - 2a - 7 > 0$ if

(i) coefficient of $x^2 > 0$ and (ii) $D < 0$

$$D = 4(4a-1)^2 - 4(15a^2 - 2a - 7) < 0$$

$$\Rightarrow 16a^2 + 1 - 8a - 15a^2 + 2a + 7 < 0$$

$$\Rightarrow a^2 - 6a + 8 < 0$$

$$\Rightarrow (a - 2)(a - 4) < 0$$

$\Rightarrow 2 < a < 4$. The only integer satisfying the condition is 3 and hence number of favourable values of $a = 1$.

Hence the required probability = $\frac{1}{89}$.

70. D

Sol. We have $x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \infty = \frac{x^2}{1 - \left(\frac{-x^2}{2}\right)} = \frac{2x^2}{2 + x^2}$

and $x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \infty = \frac{x}{1 - \left(\frac{-x}{2}\right)} = \frac{2x}{2 + x}$.

The given equation becomes

$$\sin^{-1}\left(\frac{2x^2}{2 + x^2}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{2x}{2 + x}\right)$$

$$\Rightarrow \frac{2x^2}{2 + x^2} = \frac{2x}{2 + x} \Rightarrow x = 1$$

$$\Rightarrow \cot^{-1}(x) \Rightarrow \cot^{-1}(1) = \frac{\pi}{4}.$$

71. D

Sol. $y^2 + \left(\frac{dy}{dx}\right)^2 x^2 - 2xy\left(\frac{dy}{dx}\right) - \left(\frac{dy}{dx}\right)^2 = 0$.

$$\Rightarrow y^2 + p^2 x^2 - 2xy p - p^2 = 0 \quad \left(\text{where } \frac{dy}{dx} = p\right)$$

$$\Rightarrow (y - px)^2 - p^2 = 0$$

$$\Rightarrow (y - px + p)(y - px - p) = 0$$

each of $y = px - p$ and $y = px + p$ is a Clairaut's equation.

Thus the solution is

$$(y - cx + c)(y - cx - c) = 0$$

$$\Rightarrow y = x \pm 1 \text{ (where } c = 1\text{)}.$$

72. A

Sol. $f(x) = (x - 1)(x - 2)^2(x - 3)$

$$\therefore f'(2) = 0.$$

73. B

Sol. ${}^{69}C_{3r-1} + {}^{69}C_{3r} = {}^{69}C_{r^2-1} + {}^{69}C_{r^2}$

$$\Rightarrow {}^{70}C_{3r} = {}^{70}C_{r^2}$$

$$\text{Thus } r^2 = 3r \text{ or } 70 - 3r = r^2$$

$$\text{so that } r = 0, 3 \text{ or } 7, -10.$$

Hence $r = 3$ and 7 (as the given is not defined for $r = 0$ and -10).

74. D

Sol. Given function is $f(x) = x^4(1-x)^2$, $x \in [0, 1]$
 $f(x)$ being polynomial, is continuous
 and $f'(x) = 4x^3 + 6x^5 - 10x^4$
 so $f'(x)$ exist in $(0, 1)$
 and $f(0) = 0 = f(1)$.
 $\Rightarrow$ Rolles theorem is applicable.
 Hence there exist atleast one $c \in (0, 1)$ such that $f'(c) = 0$
 $\Rightarrow 4c^3 + 6c^5 - 10c^4 = 0 \Rightarrow 2c^3(2 + 3c^2 - 5c) = 0$
 $\Rightarrow c = 0, 1, 2/3$
 $2/3 \in (0, 1)$.
 Hence all (A), (B) and (C) are correct but (D) is false.

75. A

Sol. Period of $|\sin 4x| = \frac{2\pi}{4} \times \frac{1}{2} = \frac{\pi}{4}$

Period of $|\cos 2x| = \frac{8\pi}{4} \times \frac{1}{2} = \frac{\pi}{4}$

$$\ell \text{cm} \left(\frac{\pi}{4}, \frac{\pi}{2} \right) = \frac{\pi}{2}.$$

76. D

Sol. We know that $\text{Adj}(\text{Adj } A) = |A|^{n-2} A$ if $|A| \neq 0$ [If order of $A = n$]

$\Rightarrow \text{Adj}(\text{Adj } A) = |A| \cdot A$ [Here $n = 3$]

$\Rightarrow \det \text{Adj}(\text{Adj } A) = |A|^3 \cdot \det A = |A|^4 = (14)^4$ (given)

$\Rightarrow |A| = 14$.

Now $|A| = x(1+2) - 2(-1-4) - 1(1-2) = 14$

$\Rightarrow 3x + 11 = 14 \Rightarrow x = 1$.

77. B

Sol. $f(x) = \begin{vmatrix} x+a^2 & x^4+1 & 3 \\ x+b^2 & 2x^4+2 & 3 \\ x+c^2 & 3x^4+7 & 3 \end{vmatrix}$

$$\Rightarrow f'(x) = \begin{vmatrix} 1 & x^4+1 & 3 \\ 1 & 2x^4+2 & 3 \\ 1 & 3x^4+7 & 3 \end{vmatrix} + \begin{vmatrix} x+a^2 & 4x^3 & 3 \\ x+b^2 & 8x^3 & 3 \\ x+c^2 & 12x^3 & 3 \end{vmatrix} + \begin{vmatrix} x+a^2 & x^4+1 & 0 \\ x+b^2 & 2x^4+2 & 0 \\ x+c^2 & 3x^4+7 & 0 \end{vmatrix} = 0$$

$$f'(x) = 0 + 12x^3 \begin{vmatrix} x+a^2 & 1 & 1 \\ x+b^2 & 2 & 1 \\ x+c^2 & 3 & 1 \end{vmatrix} + 0 = 0$$

$$f'(x) = \begin{vmatrix} x+a^2 & 1 & 1 \\ x+b^2 & 2 & 1 \\ x+c^2 & 3 & 1 \end{vmatrix} = 0 \quad (\text{Since } x \neq 0)$$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1$$

$$f'(x) = \begin{vmatrix} x+a^2 & 1 & 1 \\ b^2-a^2 & 1 & 0 \\ c^2-b^2 & 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow b^2 - a^2 - c^2 + b^2 = 0$$

$$\Rightarrow 2b^2 = a^2 + c^2$$

$$\Rightarrow a^2, b^2, c^2 \text{ are in A.P.}$$

78. B

Sol. The lines are $x = ay + b$; $z = cy + d$

$$x = a_1y + b_1; z = c_1y + d_1$$

$\Rightarrow$ the given equations can be written as

$$\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c} \quad \dots (1)$$

$$\text{and } \frac{x-b_1}{a_1} = \frac{y}{1} = \frac{z-d_1}{c_1} \quad \dots (2)$$

The line (1) and (2) will be perpendicular if

$$aa_1 + 1 + cc_1 = 0 \Rightarrow aa_1 + cc_1 = -1.$$

79. C

$$\text{Sol. } \sum_{r=2}^{\infty} (\tan^{-1}(r-2) - \tan^{-1}(r-3))$$

$$= \tan^{-1}0 - \tan^{-1}(-1)$$

$$\tan^{-1}1 - \tan^{-1}0$$

$$\tan^{-1}2 - \tan^{-1}1$$

$$\vdots$$

$$\tan^{-1}(n-2) - \tan^{-1}(n-3)$$

$$S_n = \tan^{-1}(n-2) + \frac{\pi}{4}$$

$$\therefore S_{\infty} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}.$$

80. A

$$\text{Sol. } \cos^2 x = \sin^2 x$$

$$\Rightarrow 1 - \sin^2 x = \sin^2 x \Rightarrow 2\sin^2 x = 1$$

$$\sin x = \pm \frac{1}{\sqrt{2}}$$

but $\sin x$ cannot be -ve.

$$\sin x = \frac{1}{\sqrt{2}} \Rightarrow x = n\pi + (-1)^n \frac{\pi}{4}$$

SECTION - B

81. 00336.00

$$\text{Sol. } x_k \geq 0, k = 0, 1, 2, 3, 4, 5$$

and

$$x_1 + x_2 + x_3 + x_4 + x_5 = 20 \text{ and } x_1 + x_2 + x_3 = 5$$

$$\Rightarrow x_4 + x_5 = 20 - 5$$

$$\Rightarrow x_4 + x_5 = 15$$

No. of integral solution

$$\binom{15+2-1}{2-1} = \binom{16}{1} = 16 \text{ and number of integral solution of } x_1 + x_2 + x_3 = 5 \text{ is}$$

$$\binom{5+3-1}{3-1} = \binom{7}{2} = \frac{7 \times 6}{2} = 21$$

Hence total number of integral solution is $21 \times 16 = 336$

82. 00000.50

Sol. $\cos \theta + \cos 3\theta + \cos 5\theta = \frac{1}{2}$

When $\theta = \pi/7$

83. 00045.00

Sol. $\int_{10}^{100} (x - [x]) dx$
 $= \int_{10}^{100} \{x\} dx$
 $= 90 \int_{10}^{11} \{x\} dx$ as $\{x\}$ is periodic with period 1
 $= 90 \times \frac{1}{2} = 45.$

84. 00003.50

Sol. $S = 29 \binom{30}{0} + 28 \binom{30}{1} + 27 \binom{30}{2} + \dots + 1 \binom{30}{28} + 0 \binom{30}{29} - \binom{30}{30}$
 $\therefore 2S = 28 \binom{30}{0} + \binom{30}{1} + \binom{30}{2} + \dots + \binom{30}{30} = 28(2^{30})$
Hence, $S = 14 \times 2^{30} = \frac{7}{2} \times 2^{32} \Rightarrow k = \frac{7}{2}$

85. 00002.00

Sol. Since, line is tangent to the circle. ; Hence, $\frac{r}{\sqrt{2}} = \sqrt{r} \Rightarrow r = 2$

86. 00019.00

Sol. Given,

$$|\alpha - \beta|^2 = 1 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 1 \Rightarrow \frac{15}{4} = \alpha\beta$$

So, $\frac{p}{q} = \frac{15}{4}; \therefore p = 15, q = 4; \text{ Hence, } p + q = 19.$

87. 00002.00

Sol. $\sum_{r=0}^n \left(\frac{r^3 + 2r^2 + 3r + 2}{(r+1)^2} \right) {}^nC_r = \sum_{r=0}^n \left(\frac{r^2 + r + 2}{r+1} \right) {}^nC_r = \sum_{r=0}^n r \cdot {}^nC_r + 2 \sum_{r=0}^n \frac{1}{r+1} \cdot {}^nC_r$

$$\begin{aligned}
 &= n \cdot 2^{n-1} + 2 \left(\frac{2^{n+1} - 1}{n+1} \right) = \frac{n \cdot 2^{n-1} + n^2 \cdot 2^{n-1} + 2^{n+2} - 2}{n+1} \\
 &\Rightarrow \frac{n \cdot 2^{n-1} + n^2 \cdot 2^{n-1} + 2^{n+2} - 2}{n+1} = \frac{2^2 + 2^3 + 2^4 - 2}{2+1} \quad (\text{given}) \\
 &\Rightarrow n = 2.
 \end{aligned}$$

88. 00000.75

Sol. Here $\frac{x}{4} = \cos t + \sin t$... (1)

and $\frac{y}{\sqrt{7}} = \cos t - \sin t$... (2)

Squaring and adding equation (1) and (2)

$$\frac{x^2}{16} + \frac{y^2}{7} = 2 (\sin^2 t + \cos^2 t)$$

$$\Rightarrow \frac{x^2}{32} + \frac{y^2}{14} = 1, \text{ which is an ellipse}$$

$$\text{eccentricity } e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{14}{32}} = \frac{3}{4}.$$

89. 00007.50

Sol. Given $\Delta OBC = 12$

$$\begin{vmatrix} 0 & 0 & 1 \\ a & 0 & 1 \\ \frac{a}{2} + 1 & \frac{a}{2} - 1 & 1 \end{vmatrix} = \pm 24$$

$$a \left(\frac{a}{2} - 1 \right) = \pm 24$$

$$\Rightarrow a(a - 2) = \pm 48$$

$$\Rightarrow a^2 - 2a \pm 48 = 0$$

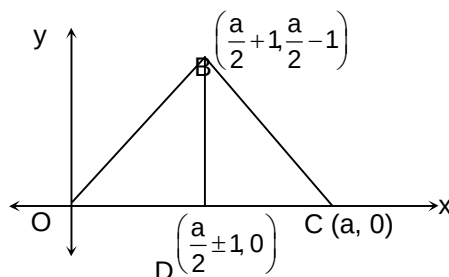
(+) sign reject

$$\therefore a^2 - 2a - 48 = 0$$

$$\Rightarrow (a - 8)(a + 6) = 0$$

$$\Rightarrow a = 8 \text{ or } a = -6 \text{ (reject)}$$

$$\therefore \text{Area of } \Delta OBD = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 5 & 3 & 1 \\ 5 & 0 & 1 \end{vmatrix} = \frac{1}{2} (15) = \frac{15}{2}$$



90. 00009.00

Sol. $\vec{v} = \frac{\lambda}{9} (\hat{i} - 7\hat{j} + 2\hat{k})$

$$|\vec{v}| = \sqrt{6} = \frac{3\sqrt{6}}{9} \Rightarrow \lambda = 3$$

$$\therefore \vec{v} \cdot \vec{a} = 9$$